Candidate Local LLMs for the Nmap-to-CVE Hallucination Study

MASTER 2027, Track 1 (AI) | Model shortlist and rationale | November 2026

1. Purpose and selection criteria

This note proposes five small language models to run locally through Ollama in the study "Can Small Local LLMs Map Nmap Results to CVEs Without Hallucinating?". Each model receives the Nmap service-detection result for a container with a known version and proposes CVE identifiers, which are then checked against the National Vulnerability Database (NVD).

Selection criteria:

- **Fits the test hardware.** Intel Core i5-11300H, 16 GB RAM, NVIDIA RTX 3050 Laptop GPU with 4 GB VRAM. Models whose default 4-bit download is roughly 2 to 3.5 GB can run mostly or fully on the GPU. 7B to 8B models (about 4.5 to 5 GB) would spill into system RAM and run noticeably slower, so they are kept as an optional extra.
- **Different developers and training approaches.** Five labs, so that differences in hallucination can be discussed in terms of model design rather than a single family.
- **Open weights available on Ollama,** so the experiment is reproducible with one pull command per model.
- **Reliable structured output.** The pipeline asks for a fixed JSON format (CVE, confidence, or "not sure"), which Ollama can enforce with a schema.
- **A spread of generations.** From a late-2024 baseline to a model released in 2026.

2. Summary

Model (Ollama tag)	Developer	Params	Size	Context	License	Distinctive trait
llama3.2:3b	Meta	3B	~2.0 GB	128K	Llama 3.2 Community	Widely used small baseline; built for edge devices
phi4-mini	Microsoft	3.8B	~2.5 GB	128K	MIT	Curated/synthetic training data; strong reasoning for its size
gemma3:4b	Google	4B	~3.3 GB	128K	Gemma Terms of Use	Multimodal; broad multilingual coverage (140 languages)
qwen3.5:4b	Alibaba	4B	~3.4 GB	256K	Apache 2.0	Newest generation (Mar 2026); optional thinking mode
granite4.2:3b	IBM	3B	~2.2 GB	128K	Apache 2.0	Enterprise-oriented; structured JSON output; thinking mode

Sizes are approximate default downloads reported by the sources listed in Section 6. Confirm with "ollama list" after pulling.

3. Model profiles

3.1 Llama 3.2 3B (Meta)

Role in the study: baseline anchor

- **Strengths:** Smallest model in the set at about 2 GB, so it is fully GPU-resident and fast. 128K context. It is the most commonly used small open model, which makes results easy to compare with other work.
- **What makes it different:** According to Meta, the 1B and 3B models were produced by pruning and distilling from larger Llama 3.1 models. Meta lists a late-2023 knowledge cutoff, so it has no knowledge of the most recent vulnerabilities.

- **Why we include it:** A familiar reference point. Its older knowledge cutoff also tests whether a model admits uncertainty on software or CVEs it cannot have seen.
- **Watch for:** General-purpose training may lead it to cite famous CVEs from memory regardless of the scanned version.

3.2 Phi-4-mini (Microsoft, 3.8B)

Role in the study: contrasting training philosophy

- **Strengths:** Strong reasoning and instruction following for its size at about 2.5 GB. One 2026 roundup reports it approaching Llama 3.1 8B on MMLU with roughly 40% less memory.
- **What makes it different:** The Phi family is trained on heavily curated and synthetic data, trading broad factual recall for reasoning ability.
- **Why we include it:** This gives a testable hypothesis: a model that stores fewer specific facts may know fewer CVE identifiers. The study can measure whether that leads to more abstentions ("not sure") or more fabricated CVEs.
- **Watch for:** Confident-sounding answers built from reasoning rather than recalled facts.

3.3 Gemma 3 4B (Google)

Role in the study: third major lab, multilingual and multimodal design

- **Strengths:** About 3.3 GB with a 128K context window. Multimodal at this size and covers about 140 languages. Quantization-aware-trained variants exist.
- **What makes it different:** A different lab, tokenizer and training recipe from the other models. Its tool calling is reported to be weaker than Llama or Mistral, which does not matter here because the pipeline uses plain JSON output rather than tool calls.
- **Why we include it:** Adds a Google model so results are not limited to one or two ecosystems.
- **Watch for:** Together with Qwen it has the largest GPU footprint in the set. Use a small context window (4,096 tokens) so it stays on the GPU. Gemma 4 is the newer generation, but its smallest default download (about 7.2 GB) exceeds 4 GB of VRAM, so it is left for future work.

3.4 Qwen3.5 4B (Alibaba)

Role in the study: newest generation and thinking-mode ablation

- **Strengths:** Released in March 2026, about 3.4 GB, 256K context, natively multimodal, with tools and a thinking mode. The Qwen3.5 family uses a hybrid architecture (Gated Delta Networks combined with sparse mixture-of-experts).
- **What makes it different:** It is the only model in the set with a reasoning ("thinking") mode on by default. One benchmark on a 16 GB MacBook measured the 4B version using 1,457 tokens and 62 seconds for a simple question with thinking on, against 426 tokens and 17 seconds with it off, with the same correct answer.
- **Why we include it:** A current-generation representative from a fourth lab. It also allows a clean extra research question: does thinking reduce hallucination, and what does it cost in latency?
- **Watch for:** Turn thinking off in the main runs so timings are comparable, then run a smaller ablation with it on.

3.5 Granite 4.2 3B (IBM)

Role in the study: enterprise-oriented design

- **Strengths:** About 2.2 GB, 128K context, Apache 2.0 license. Described as a dense family with a thinking mode and structured JSON output, aimed at enterprise tasks such as retrieval and tool use.
- **What makes it different:** Designed with enterprise reliability in mind and with a permissive license, which matters for security teams that want to deploy models in-house.

- **Why we include it:** Tests whether a model tuned for structured, business-style output is more conservative, for example by abstaining more often, than general-purpose chat models.
- **Watch for:** Confirm the exact tag exists in the Ollama library before pulling, because Granite tags have been updated frequently.

4. Settings for a fair comparison

- Record the Ollama version and each model's digest (ID column of "ollama list") in the paper, because tags can be updated.
- Use each model's default Ollama quantization, and say so. Do not mix quantization levels between models.
- Use the same context length for all models (4,096 tokens is enough for an Nmap result plus a CVE shortlist) and the same prompts and JSON schema.
- Run the main accuracy comparison at temperature 0 with a fixed seed. Measure answer stability separately at a non-zero temperature (for example 0.7), because at temperature 0 repeated runs are almost identical by design.
- Disable thinking in the main runs. Report thinking mode as a separate ablation for the models that support it.
- Check with "ollama ps" that each model runs 100% on the GPU. If a model is split between GPU and CPU, report it, because it changes the response time.
- Load one model at a time, make a warm-up call before timing, and close the Docker lab during LLM runs to free RAM.

5. Optional additions

- **llama3.1:8b** (about 4.9 GB) as a size reference to show whether a larger model hallucinates less. It will only partly fit in 4 GB of VRAM, so it is slower.
- **gemma4:e2b-it-qat** (about 4.3 GB) as a newer-generation Google comparison, if it runs acceptably on the available GPU.

Pull commands for the five main models (confirm names on ollama.com/library first, because tags change quickly):

```
ollama pull llama3.2:3b
ollama pull phi4-mini
ollama pull gemma3:4b
ollama pull qwen3.5:4b
ollama pull granite4.2:3b
```

6. Sources

Model sizes, context windows, release dates and capabilities were taken from the following pages (accessed November 2026) and should be re-verified with the model pages on ollama.com before final reporting:

- ComputingForGeeks, "Ollama Models Cheat Sheet: Best Models by VRAM, Tested", updated 26 October 2026: computingforgeeks.com/ollama-models-cheat-sheet
- Convly, "Ollama Models List 2026": convly.ai/ollama-models-list-2026
- WebsCraft, "Best Ollama Models for 8GB RAM in 2026": webscraft.org/blog/ollama-na-8-gb-ram-yaki-modeli-pratsyuyut-u-2026
- ToolHalla, "Best Ollama Models in 2026": toolhalla.ai/blog/best-ollama-models-2026
- Gemma 4 on Ollama (tags and sizes): verdent.ai/guides/gemma-4-ollama and itechguides.com